to tell a story about many such worlds, but the postulates of the theory (wave functions, smooth evolution) don't even mention "worlds" at all. Where do the worlds come from, and why do worlds look approximately classical?

In our discussion of decoherence, we pointed out that you can think of a quantum system as having split into multiple separate copies once it becomes entangled with the larger environment around it, since whatever happens to each copy won't be able to interfere with whatever happens to the others. If we want to be sticklers, however, that's telling us that we're *allowed* to think about the decohered wave function as describing separate worlds—not that we *should* think of it that way, much less that we *need* to think of it that way. Can we do better?

The truth is, nothing forces us to think of the wave function as describing multiple worlds, even after decoherence has occurred. We could just talk about the entire wave function as a whole. It's just really helpful to split it up into worlds.

Many-Worlds describes the universe using a single mathematical object, the wave function. There are many ways of *talking about* the wave function that give us physical insight into what is going on. It may be useful in some cases to talk in terms of position, for example, and in other cases in terms of momentum. Likewise, it is often helpful to talk about the post-decoherence wave function as describing a set of distinct worlds; that's justified, because what happens on each branch doesn't affect what happens on the others. But ultimately, that language is a convenience for us, not something that the theory itself insists on. Fundamentally, the theory just cares about the wave function as a whole.

By way of an analogy, think of all the matter in the room around you right now. You could describe it—helping ourselves to the classical approximation for the moment—by listing the position and velocity of every atom in the room. But that would be crazy. You neither have access to all that information, nor could you put it to use if you did, nor do you really need it. Instead, you chunk up the stuff around you into a set of useful concepts: chairs, tables, lights, floors, and so on. That's an enormously more compact description than listing every atom would be, but still gives us a great deal of insight into what's going on.

Similarly, characterizing the quantum state in terms of multiple worlds isn't necessary—it just gives us an enormously useful handle on an